Experiment 18: Compare Contextual Embeddings for Two Semantically Similar Sentences

Python Program

```
import torch

from transformers import BertTokenizer, BertModel

from torch.nn.functional import cosine_similarity


tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")

model = BertModel.from_pretrained("bert-base-uncased")


s1 = "I love learning Python"

s2 = "I enjoy studying Python"


def get_embedding(sentence):

    inputs = tokenizer(sentence, return_tensors="pt")

    with torch.no_grad():

        output = model(**inputs)

    return output.last_hidden_state[:, 0, :]


e1 = get_embedding(s1)

e2 = get_embedding(s2)


score = cosine_similarity(e1, e2).item()


print("Sentence 1:", s1)
```

```
print("Sentence 2:", s2)

print("Cosine Similarity:", round(score, 4))
```

Output

